

London Petro Academy Ltd

Hydrogen Technology, Economics & Business Models

End-to-End Hydrogen Value Chains, Costs and Bankable
Business Cases for the Energy Transition



LONDON PETRO ACADEMY
Empowering Professionals for a Sustainable Future

2026/2027

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Course Overview

This multi-day Hydrogen: Technology, Economics & Business Models course provides a structured, end-to-end view of hydrogen across production, transport, storage and end-use applications, with a strong focus on commercial viability and investment decision-making. It begins by positioning hydrogen as a potential game-changer in 21st-century energy markets, driven by decarbonisation, ammonia and methanol production, and rising demand from fuel cell electric vehicles and flexible power systems. Participants explore current and emerging production pathways (grey, blue, turquoise and green hydrogen), the role of carbon capture and storage, and the economics of power-to-hydrogen, including CAPEX/OPEX drivers, levelised cost of hydrogen and future cost roadmaps. The course then examines hydrogen transportation, distribution and storage options, sector coupling between electricity and gas infrastructure, and the logistical and cost challenges of building scalable value chains. Finally, it delves into energy and transport applications, from industrial use and CHP to mobility and hydrogen-derived fuels, tying technology choices back to revenue stacking, risk, policy frameworks and realistic business cases.

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COURSE OBJECTIVES

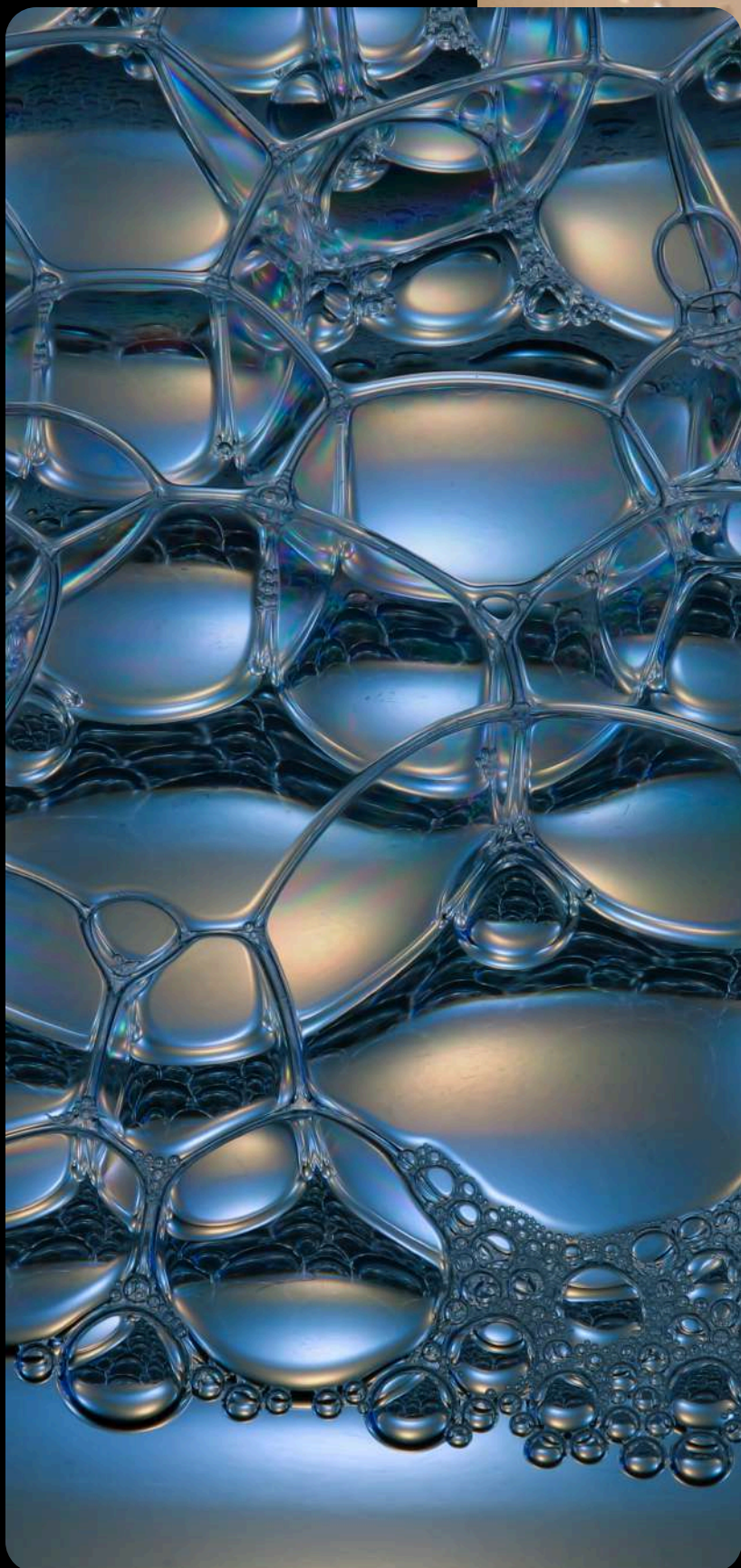
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By the end of this course, participants will understand the current status of hydrogen markets and recent developments, as well as the major drivers and inhibitors influencing growth across regions and sectors. They will be able to compare key hydrogen production technologies, distinguish between different colours of hydrogen and assess their respective cost, carbon and maturity profiles. Attendees will gain a solid grasp of the hydrogen value chain, including transmission, distribution and storage elements, and the technical and economic constraints associated with compression, liquefaction, refuelling and grid blending. They will be able to map hydrogen's many applications in petrochemicals, industry, power systems and transport, and analyse real business cases from around the world to understand economics, competing technologies and value-stacking opportunities. Participants will also gain insight into policy and regulatory frameworks that support hydrogen deployment, investment scenarios, and the potential of clean hydrogen as a new commodity and enabler of the broader energy transition.

Course audience

This course is designed for professionals across the energy, industrial and finance sectors who are evaluating hydrogen projects, strategies or investments and need a comprehensive, commercially grounded understanding of hydrogen technologies and business models. It is particularly relevant for project developers and sponsors, OEMs, EPC contractors and O&M service providers, oil and gas upstream companies, petrochemical and process industry firms, transport companies and port operators, IPPs, utilities and grid operators. Private equity and institutional investors, development finance institutions, commercial lenders, and legal, technical and insurance advisors will benefit from the detailed treatment of costs, risks and bankability. The course is also suitable for government officials and regulators, corporate procurement and contract managers, and business developers and sales managers who must interpret hydrogen's role in decarbonisation strategies and shape future market and policy frameworks.





Hydrogen production processes

- The hydrogen energy value chain from production, transport, storage and end usage
- Production pathways and their current level of maturity
- Understanding differences of green, blue, turquoise and grey hydrogen
- Natural gas steam methane reformation and coal gasification
- Carbon capture and storage and the relation to hydrogen
- Hydrogen production via electrolysis
- Renewable Hydrogen Technologies and Applications
- Drivers of levelised cost of hydrogen production
- Economics of Power-to-Hydrogen
- Technology CAPEX, OPEX and performance
- Hydrogen facility costs other than equipment costs
- Future cost trends and roadmaps
- Value to be captured from power-to-hydrogen applications

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Hydrogen transportation, distribution and storage

- Transmission, distribution and storage elements of hydrogen value chain
- Transport options for hydrogen
- Overview and requirements of hydrogen storage options
- Hydrogen compression and liquefaction
- Hydrogen refuelling stations
- Sector coupling – a united infrastructure for electricity and gas
- Limits on hydrogen blending in natural gas networks
- Feasibility of using existing transmission and distribution networks
- Logistical challenges in transporting and storing hydrogen

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Energy Applications

- Increased power system flexibility delivering electrical grid services
- Large scale renewable integration, grid constraints and hydrogen as a long-term energy storage
- Industrial applications: Refineries, Steel, Chemical industry and light industry
- Micro-CHP
- Commercial and large scale CHP
- Fuel cell system for re-electrification
- Converting hydrogen to hydrogen-based fuels and feedstocks
- Ammonia and methanol conversion

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This course is available In-House and can be tailored to your specific requirements.



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Touchpoints

For more information, please contact us on:

+44 (0) 1582 516247

Alternatively, you can email us at info@londonpetroacademy.co.uk

Question about this course?

Call us on +44 (0) 1582 516247 and one of our representatives will assist you with your enquiry.

3 Easy ways to register

Online: www.londonpetroacademy.co.uk

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